what is a research question?

prof. dr. nicolas meseth

1 from topic to question
2 what makes a good question
3 questions without an answer
4 sharpening a question
5 the reality test

a data set with 606 columns answers nothing by itself, until a question tells you which shelf to light up.

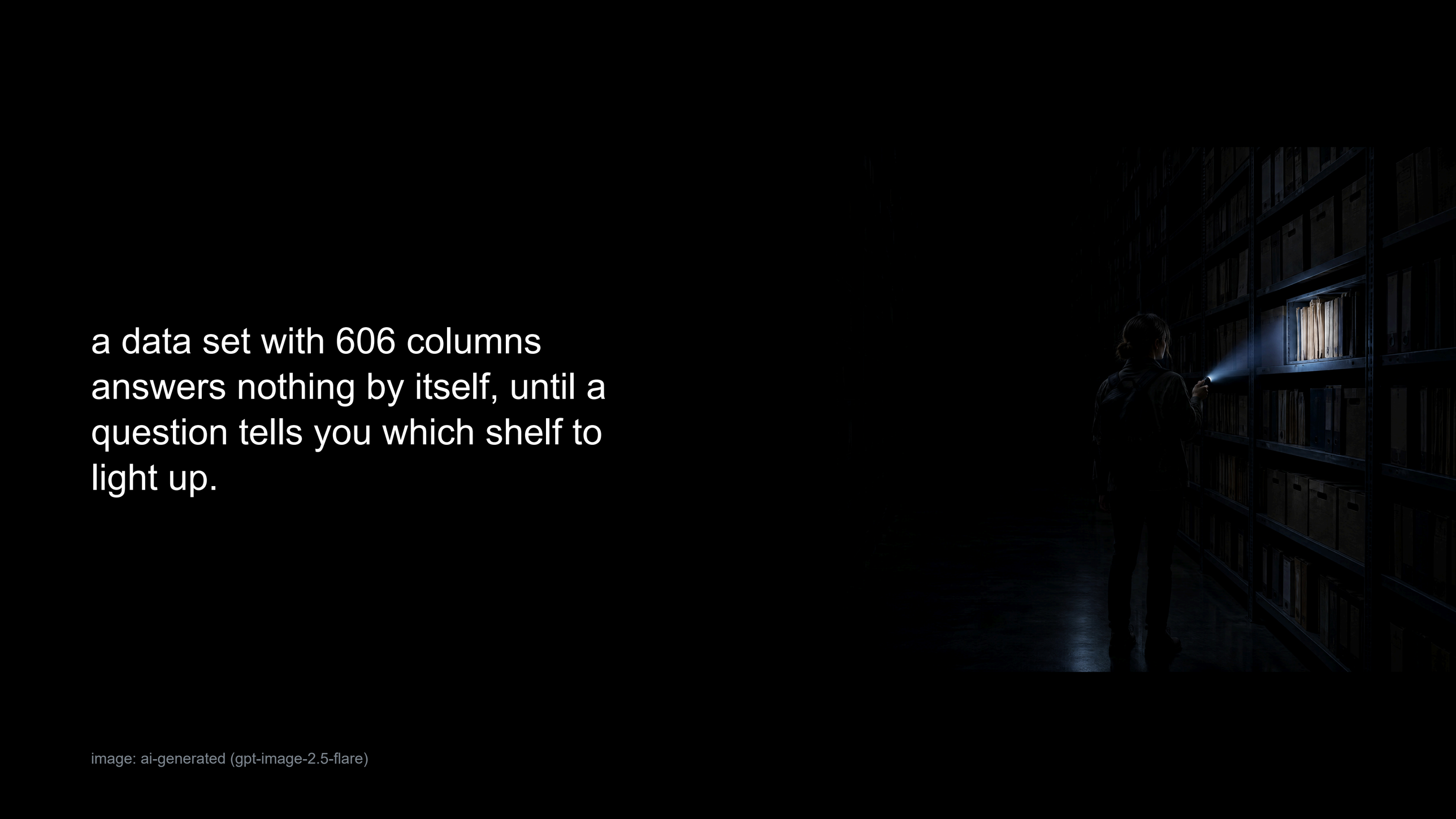


image: ai-generated (gpt-image-2.5-flare)

part 1

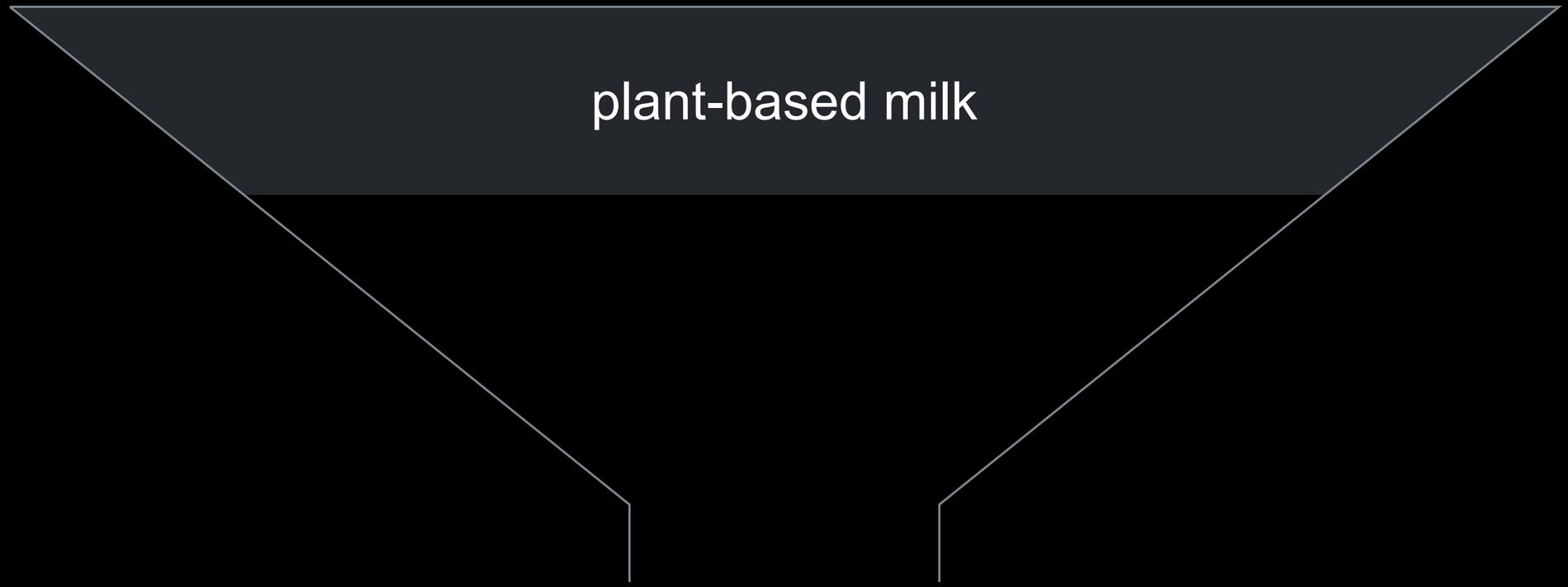
from topic to question

plant-based milk is a topic, and a topic names a shelf without saying what you want to know about it.



image: ai-generated (gpt-image-2.5-flare)

topic



after Döring (2023), Forschungsmethoden und Evaluation, p. 148



after Döring (2023), Forschungsmethoden und Evaluation, p. 148

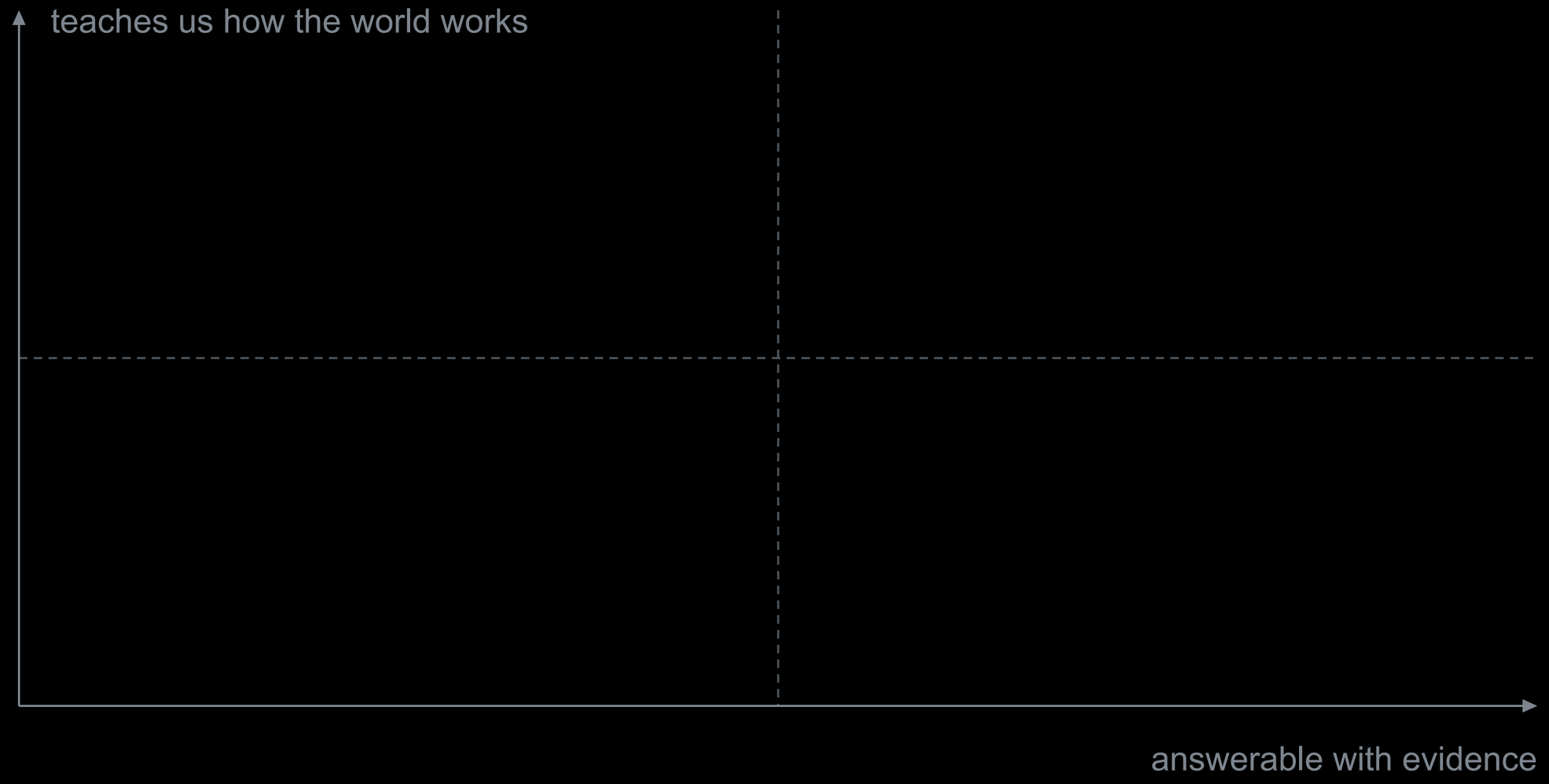


question

how does purchase certainty for alpro oat drink change when its price rises from 1.79 to 2.09 euros?

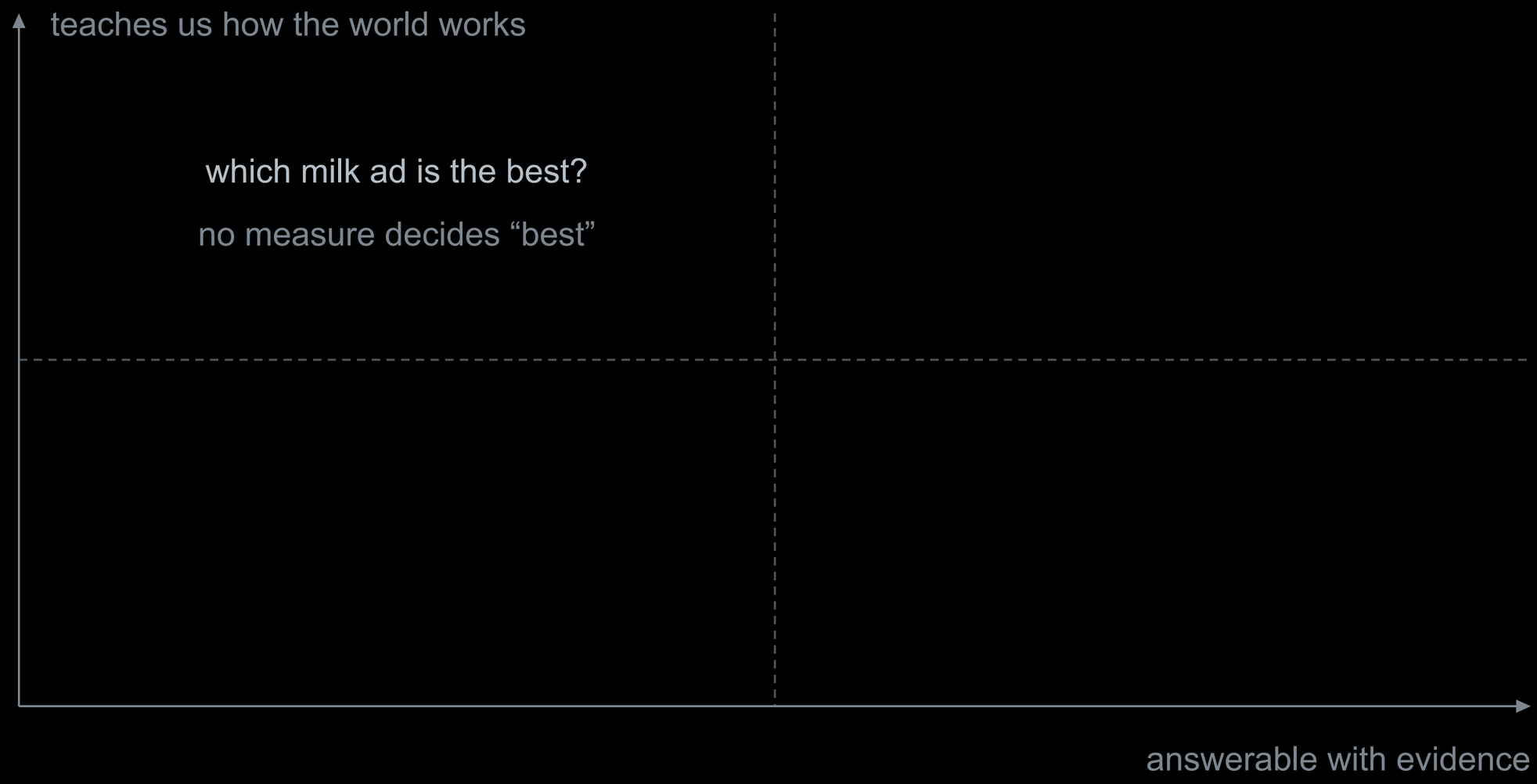
after Döring (2023), Forschungsmethoden und Evaluation, p. 148

two conditions



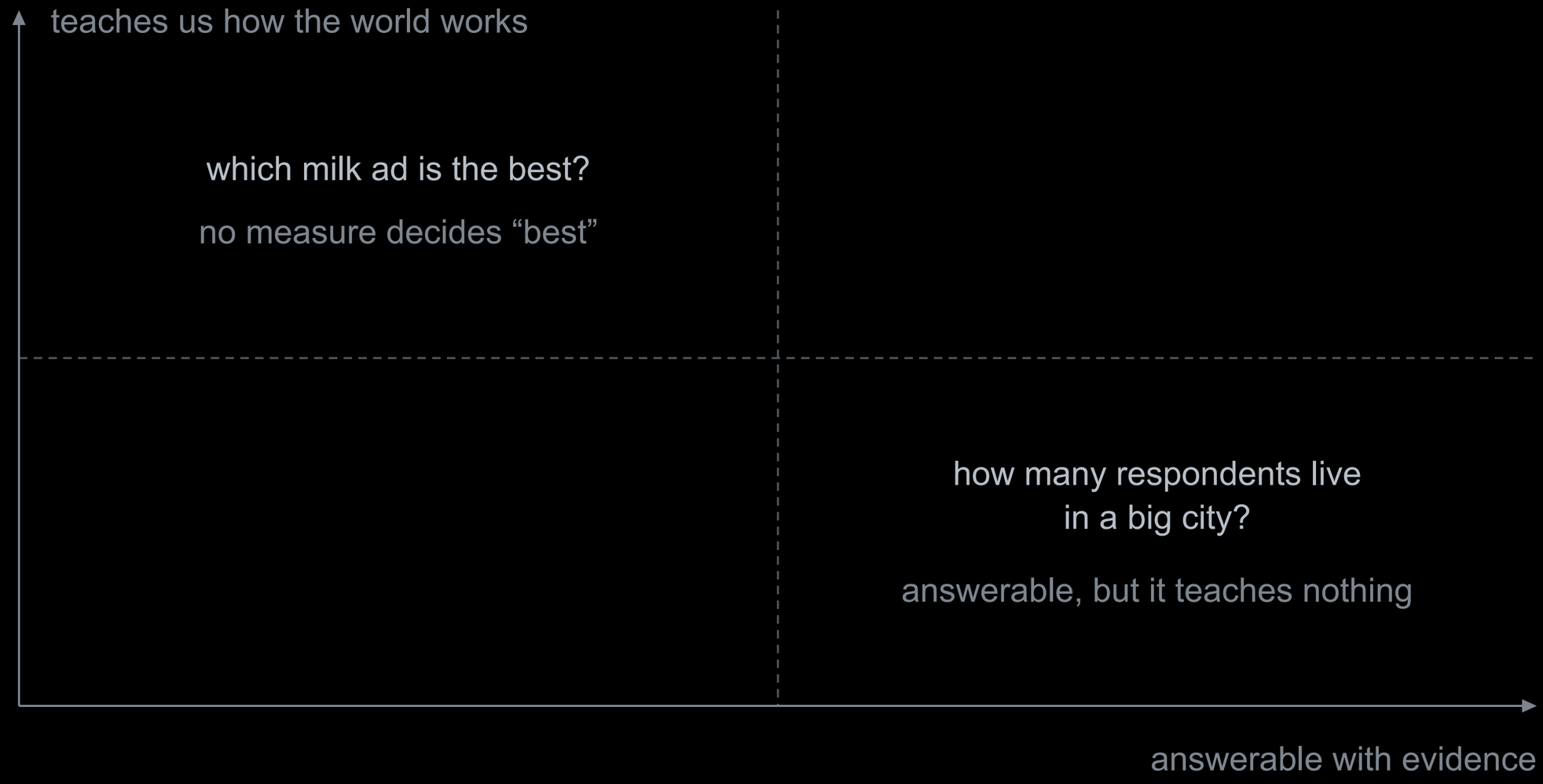
after Huntington-Klein, The Effect, ch. 2

two conditions



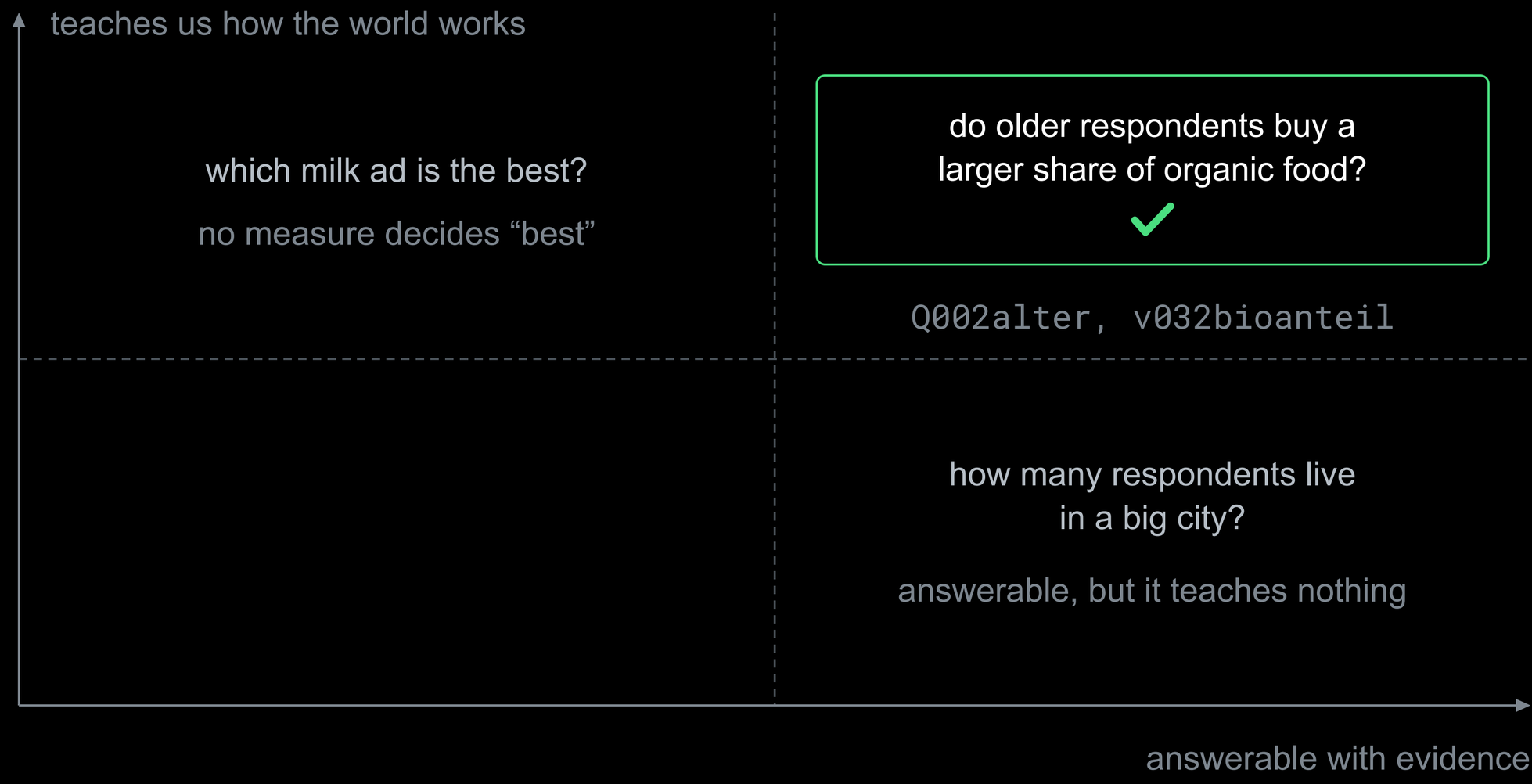
after Huntington-Klein, The Effect, ch. 2

two conditions



after Huntington-Klein, The Effect, ch. 2

two conditions



after Huntington-Klein, The Effect, ch. 2

research question

does the share of
organic food people buy
differ by age?

open: it asks, without
expecting a direction

with a survey that is already collected, the open question is the normal case.

after Döring (2023), Forschungsmethoden und Evaluation, pp. 149–152

research question

does the share of organic food people buy differ by age?

open: it asks, without expecting a direction

hypothesis

respondents over 60 buy a larger share of organic food than respondents under 30.

a claim with a direction, based on theory you cite

with a survey that is already collected, the open question is the normal case.

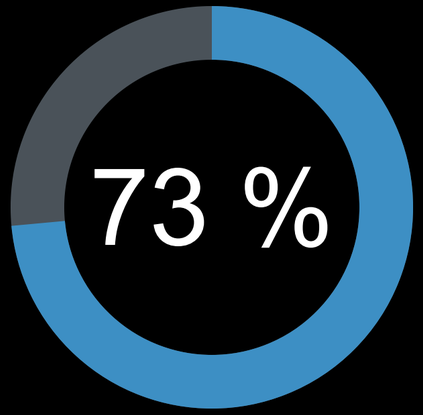
after Döring (2023), Forschungsmethoden und Evaluation, pp. 149–152

part 2

what makes a good question

descriptive

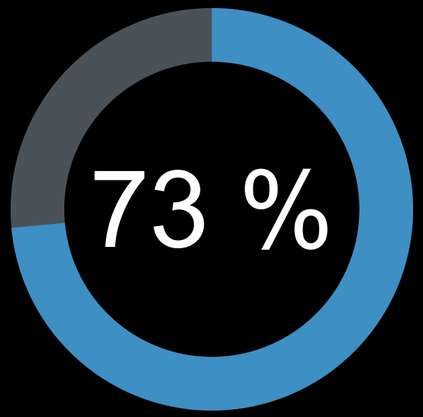
what share of the respondents shops at a discounter?



v008ort_1discount

descriptive

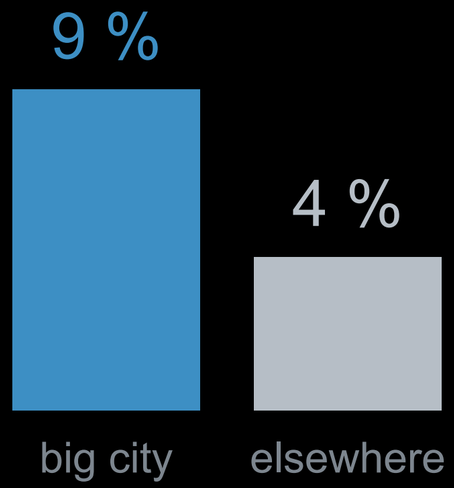
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`v008ort_1discount`

relational

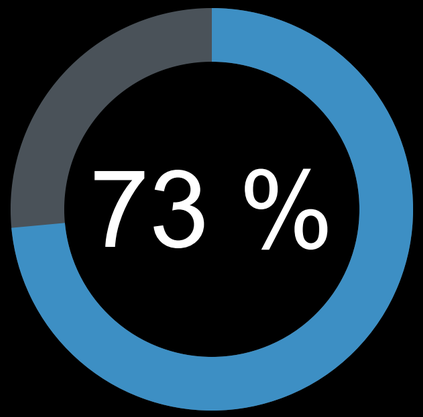
do people in big cities shop online more often?



`D044stadtf, v008ort_9online`

descriptive

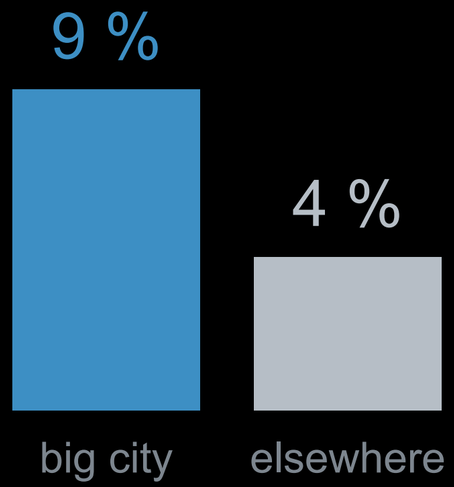
what share of the respondents shops at a discounter?



`v008ort_1discount`

relational

do people in big cities shop online more often?



`D044stadtf, v008ort_9online`

causal

does a higher price lower purchase certainty for alpro oat drink?

1.69 | 1.79 | 1.89 | 2.09

⚠

only with a design that separates the cause, here an experiment

`ad015p1reaktion4, U015p1reaktion4_3alpro`

pilots go through their checklist
before take-off, and you go through
yours before you compute anything.



image: ai-generated (gpt-image-2.5-flare)

potential results

would every possible answer teach you something?

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feasibility

do the data contain what you need?

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can you answer it in the time you have?

after Huntington-Klein, The Effect, 2.4

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can these data separate what you want to separate?

after Huntington-Klein, The Effect, 2.4

potential results

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do the data contain what you need?

scale

can you answer it in the time you have?

design

can these data separate what you want to separate?

simplicity

is it one question, not three?

after Huntington-Klein, The Effect, 2.4

too broad, just right

what drives organic food purchases?

a whole research programme, not one question

a narrow question names one variation, one outcome and one comparison, and then it has an answer.

too broad, just right

what drives organic food purchases?

a whole research programme, not one question

do vegetarians buy a larger share of organic food than people who eat meat without restrictions?

variation	`V041nofleischf`
outcome	`v032bioanteil`
comparison	vegetarians against no restrictions

a narrow question names one variation, one outcome and one comparison, and then it has an answer.

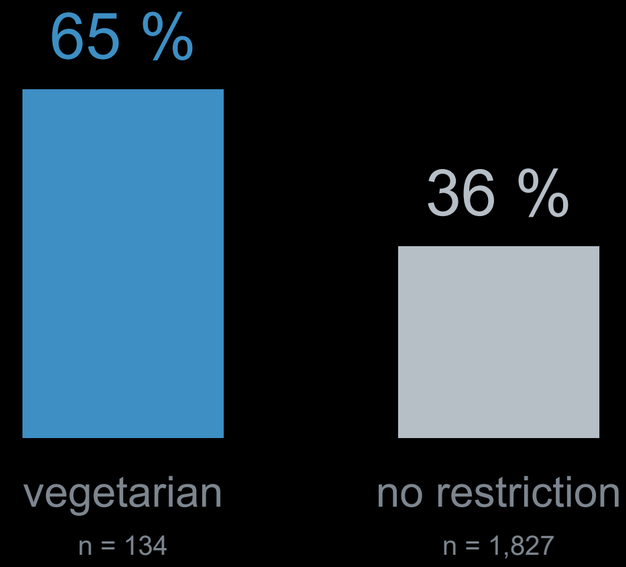
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do vegetarians buy a larger share of organic food than people who eat meat without restrictions?

variation	`V041nofleischf`
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a narrow question names one variation, one outcome and one comparison, and then it has an answer.

a question you can answer well is worth more than an important one you can only talk about.

part 3

questions without an answer

some questions sound important and still find an empty shelf in your data.



image: ai-generated (gpt-image-2.5-flare)

four traps

normative

should oat milk cost as much as cow's milk?

data describe what is,
they do not judge it

questions without an answer

four traps

normative

should oat milk cost as much as cow's milk?

data describe what is,
they do not judge it

vague

what do germans think about food?

“think about food” names
nothing you could measure

questions without an answer

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causal without design

does a higher income make people buy organic?

nobody assigned income at random,
and it goes with education and values

four traps

normative

should oat milk cost as much as cow's milk?

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vague

what do germans think about food?

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causal without design

does a higher income make people buy organic?

nobody assigned income at random,
and it goes with education and values

not in the data

has organic buying grown since 2020?

one survey at one time,
november 2025

does a higher income make people buy organic?

causal, without a design that could show it

“make” becomes “is related to”, and the report says honestly why it cannot claim more.

does a higher income make people buy organic?

causal, without a design that could show it

↓

is household income related to the share of organic food people buy?

`d049einkommen, v032bioanteil`

✓

“make” becomes “is related to”, and the report says honestly why it cannot claim more.

which question can the milk survey answer?

is milk too cheap?

do vegans use farm shops?

did prices rise since 2020?

do ads cause brand loyalty?

live.hsos.uber.space/q

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part 4

sharpening a question

a vague idea becomes a question once you can say who, where, when, what varies, what you measure and what you compare.



image: ai-generated (gpt-image-2.5-flare)

prices and
oat milk

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oat milk

who people who bought and drank milk or milk alternatives in the last 12 months

prices and
oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany

prices and
oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	november 2025

prices and oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	november 2025
variation	price of alpro oat drink: 1.69, 1.79, 1.89 or 2.09 euros

prices and oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	november 2025
variation	price of alpro oat drink: 1.69, 1.79, 1.89 or 2.09 euros
outcome	purchase certainty from −4 to +4

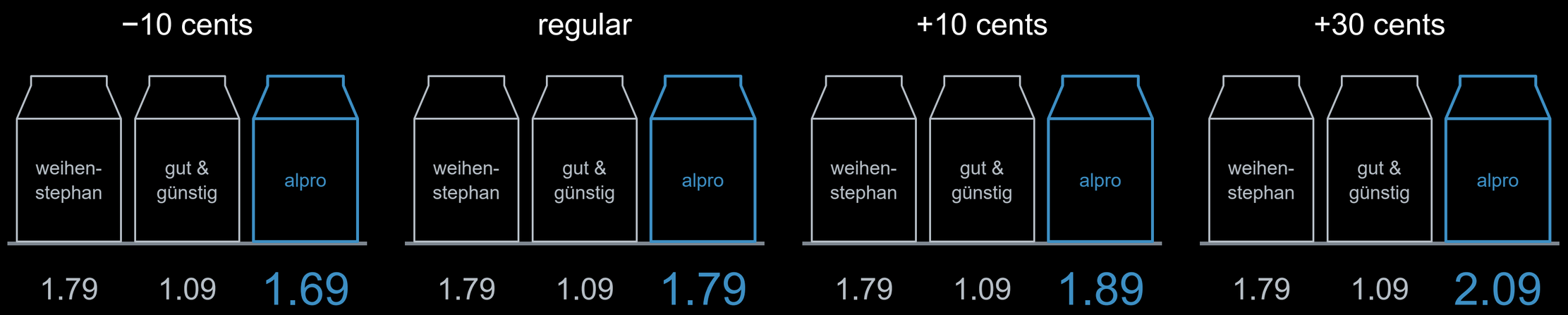
prices and oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	november 2025
variation	price of alpro oat drink: 1.69, 1.79, 1.89 or 2.09 euros
outcome	purchase certainty from −4 to +4
comparison	each price against the regular 1.79

the sharpened question

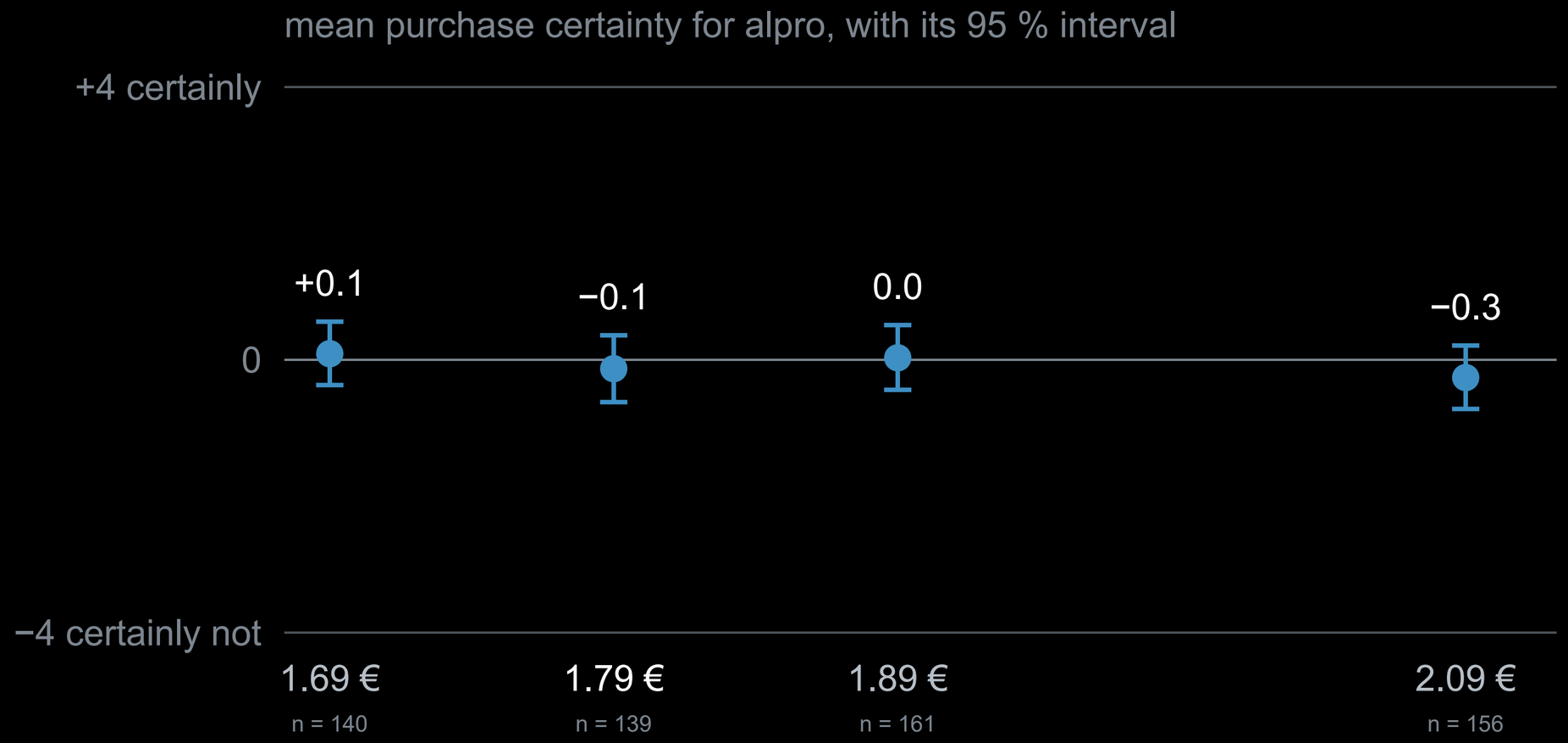
how does purchase certainty for alpro oat drink change among people who buy and drink milk when its price lies 10 cents below, or 10 or 30 cents above, the regular 1.79 euros?

chance decided which shelf each respondent saw



chance decided which shelf each respondent saw, so a difference in purchase certainty can be put down to the price.

and the answer



the price hardly moved purchase certainty, and for a good question that is an answer too.

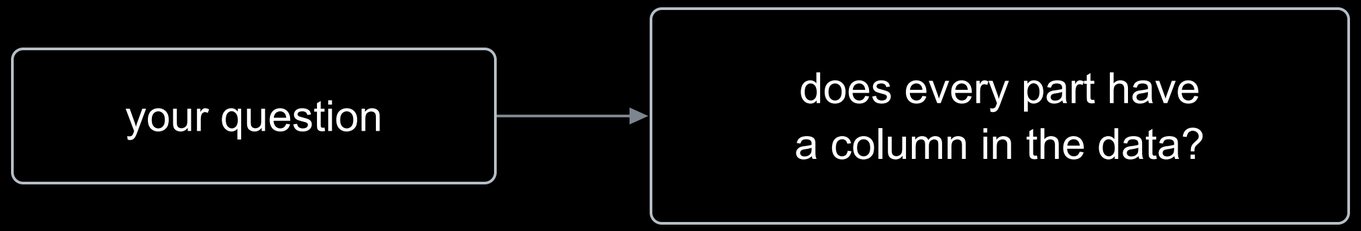
part 5

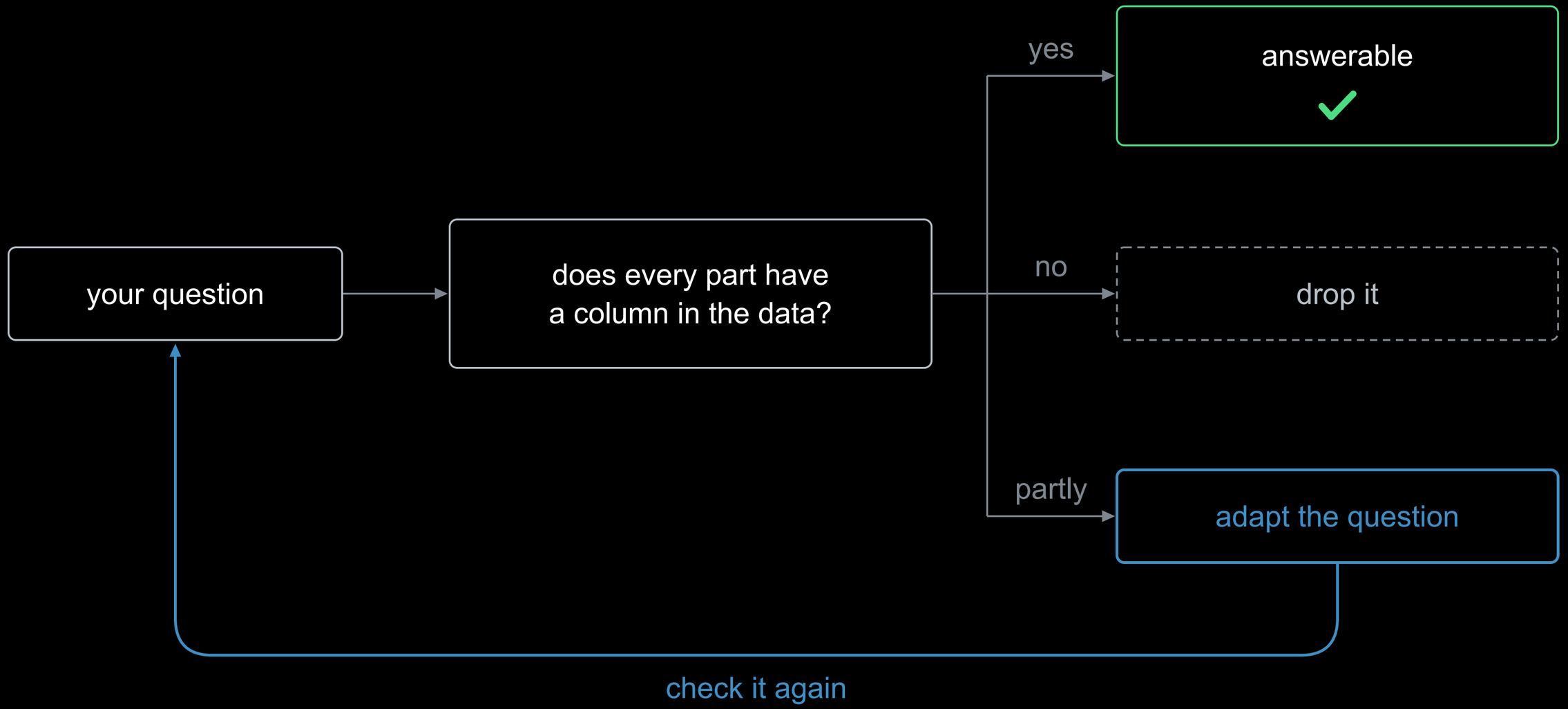
the reality test

the data came first, so the question has to fit them and not the other way round.



image: ai-generated (gpt-image-2.5-flare)


your question
does every part have
a column in the data?


your question
does every part have
a column in the data?
yes
answerable
no
drop it
partly
adapt the question
check it again

how does purchase certainty for alpro oat drink change among people who buy and drink milk when its price lies 10 cents below, or 10 or 30 cents above, the regular 1.79 euros?

in the question	column	what the codebook says
people who buy and drink milk	`v006gekauft_3mi == 3`	bought and drank milk or milk alternatives in 12 months
its price	`ad015p1reaktion4`	codes 1 to 4 for 1.69 to 2.09 €, assigned by chance
purchase certainty	`U015p1reaktion4_3alpro`	from −4 for certainly not to +4 for certainly
the regular 1.79 euros	`ad015p1reaktion4 == 2`	“Normal alpro” in AD015p1reaktion4f

601 respondents saw a price and 596 rated alpro; time and place are not columns

codebook: the documentation of the data, with every column, its question, who was asked and what each code means

write your question down before you compute anything, check every word against the codebook, and change the question rather than the data.



image: ai-generated (gpt-image-2.5-flare)